DATASHEET

5mm photodiode
PD333-3B/H0/L2

Features

- Fast response time
- High photo sensitivity
- Small junction capacitance
- Pb free
- The product itself will remain within RoHS compliant version
- Compliance with EU REACH

Description

- PD333-3B/H0/L2 is a high speed and high sensitive PIN photodiode in a standard 5Φ plastic package. Due to its black epoxy the device is sensitive to infrared radiation

Applications

- High speed photo detector
- Security system
- Camera

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Device Selection Guide

Chip Materials	Lens Color
Silicon	Black

Absolute Maximum Ratings (Ta=25℃)

Parameter	Symbol	Rating	Unit
Reverse Voltage	V_R	32	V
Operating Temperature	T_{opr}	-25 ~ +85	℃
Storage Temperature	T_{stg}	-40 ~ +100	℃
Soldering Temperature	T_{sol}	260	℃
Power Dissipation at (or below) 25℃ Free Air Temperature	P_c	150	mW

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Electro-Optical Characteristics (Ta=25℃)

Parameter	Symbol	Min.	Typ.	Max.	Unit	Condition
Range Of Spectral Bandwidth	$\lambda_{0.5}$	840	-----	1100	nm	-----
Wavelength Of Peak Sensitivity	λ_P	-----	940	-----	nm	-----
Open-Circuit Voltage	V_{OC}	-----	0.39	-----	V	Ee=1mW/cm^2 λp=940nm
Short- Circuit Current	I_{SC}	-----	35	-----	μA	Ee=1mW/cm^2 λp=940nm
Reverse Light Current	I_L	25	35	-----	μA	Ee=1mW/cm^2 λp=940nm V_R=5V
Reverse Dark Current	I_D	----	5	30	nA	Ee=0mW/cm^2 V_R=10V
Reverse Breakdown Voltage	V_{BR}	32	170	-----	V	Ee=0mW/cm^2 I_R=100μA
Total Capacitance	C_t	-----	18	-----	pF	Ee=0mW/cm^2 V_R=5V f=1MHz
Rise Time/ Fall Time	t_r / t_f	-----	45	-----	ns	V_R=10V R_L=100Ω
View Angle	2θ1/2	-----	80	-----	deg	I_F=20mA

Note:
Tolerance of Dominant Wavelength: ±1nm
Tolerance of Forward Voltage: ±0.1V

I_L Rank

Condition：Ee=1mW/cm^2、λp=940nm、V_R=5V
Unit：uA

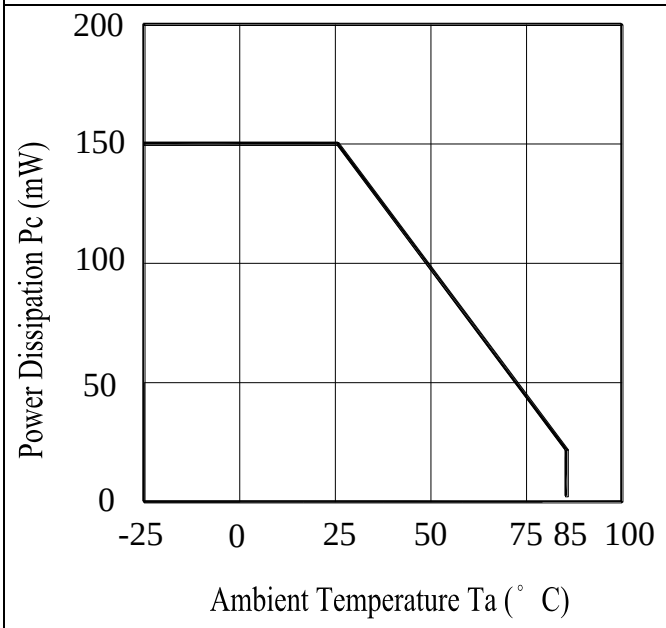
Bin Number	BIN1	BIN2	BIN3	BIN4	BIN5
Min	25	35	45	55	65
Max	35	45	55	65	75

Tolerances： 20%

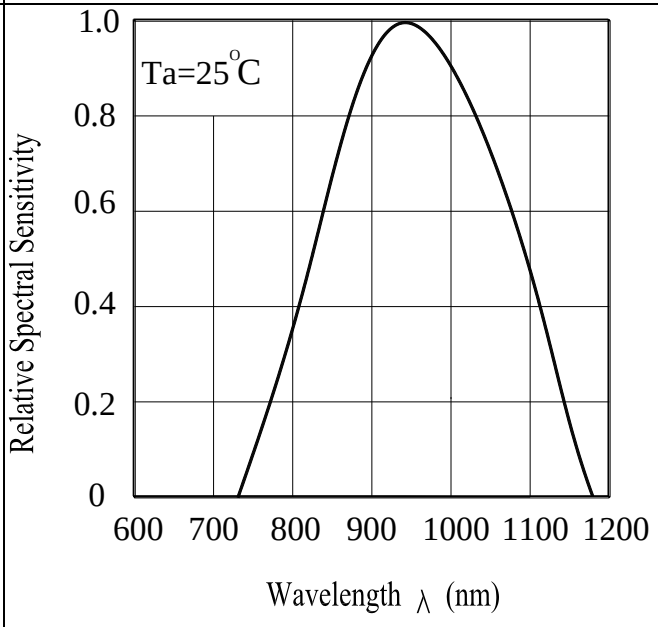
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Typical Electro-Optical Characteristics Curves

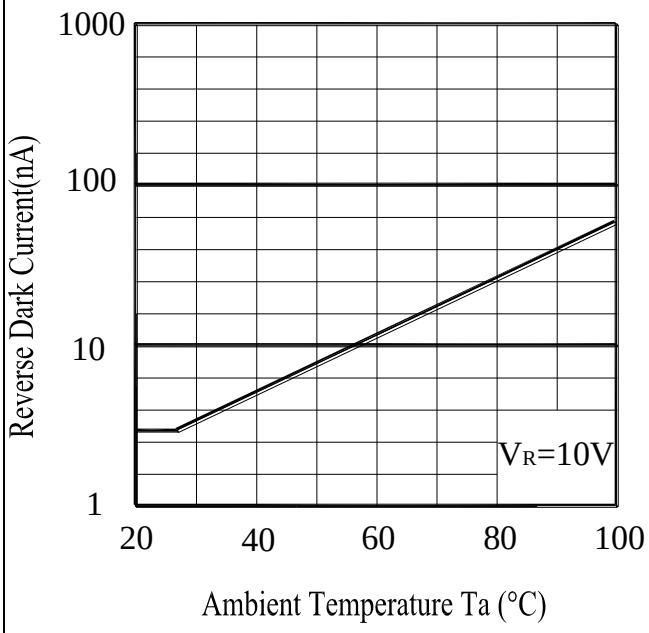
Power Dissipation vs. Ambient Temperature



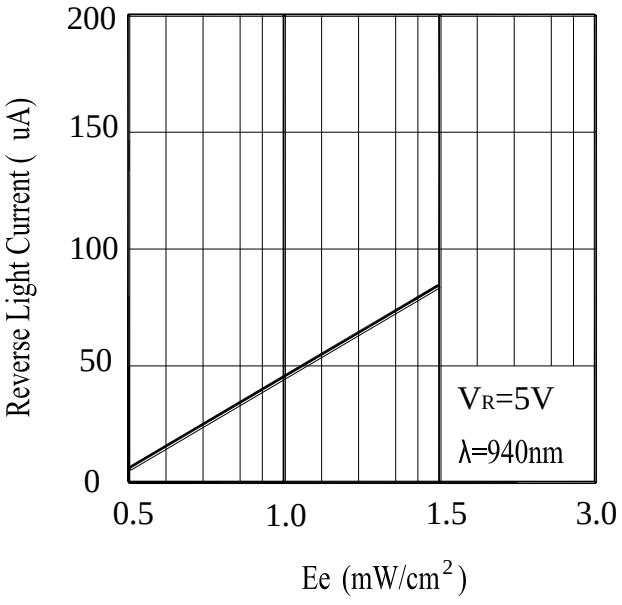
Spectral Sensitivit



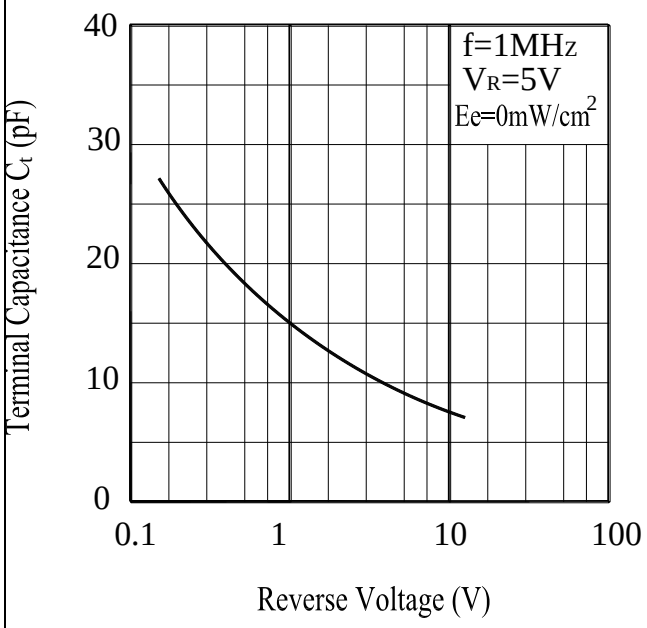
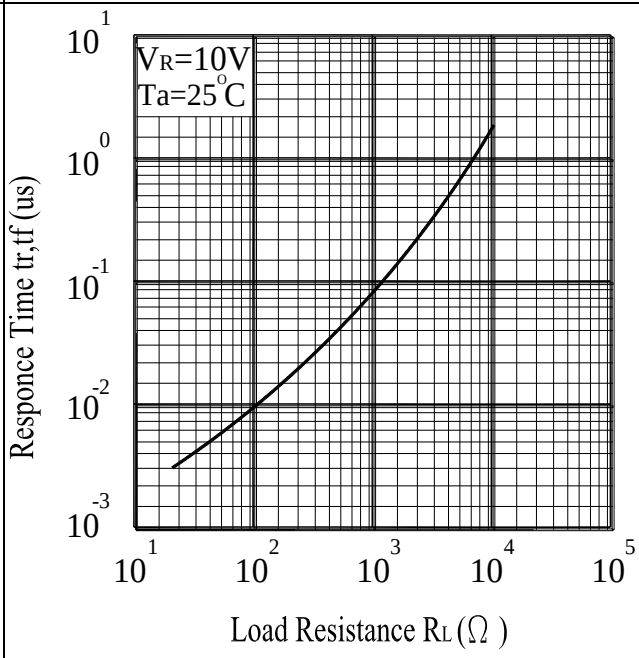
Reverse Dark Current vs. Ambient Temperature



Reverse Light Current vs. Ee

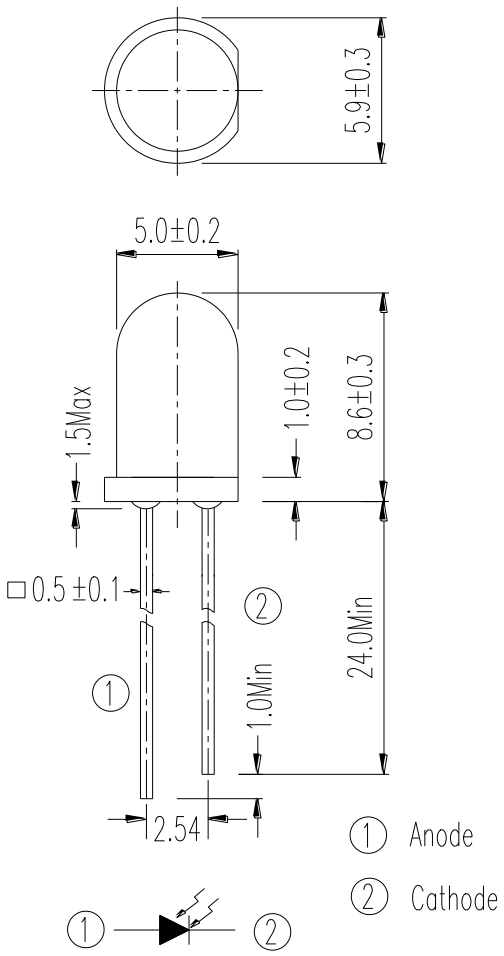


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Terminal Capacitance vs. Reverse Voltage	Response Time vs. Load Resistance
	

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Package Dimension

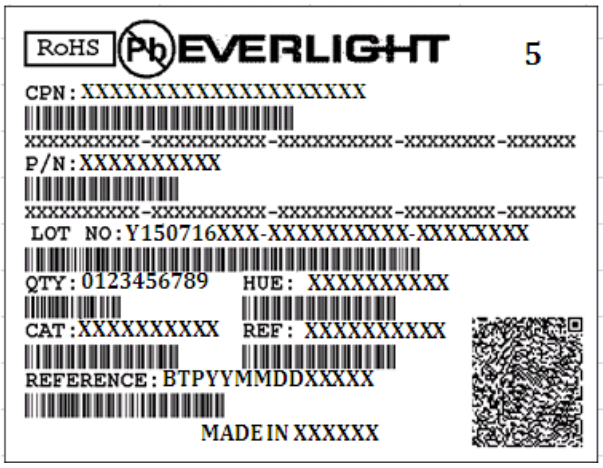


Note: Tolerances unless dimensions ±0.25mm

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Packing Quantity Specification
1.200~500PCS/1Bag，5Bags/1Box
2.10Boxes/1Carton

Label Form Specification



- CPN: Customer's Product Number
- P/N: Product Number
- QTY: Packing Quantity
- CAT: Luminous Intensity Rank
- HUE: Dom. Wavelength Rank
- REF: Forward Voltage Rank
- LOT No: Lot Number
- X: Month
- Reference: Identify Label Number

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